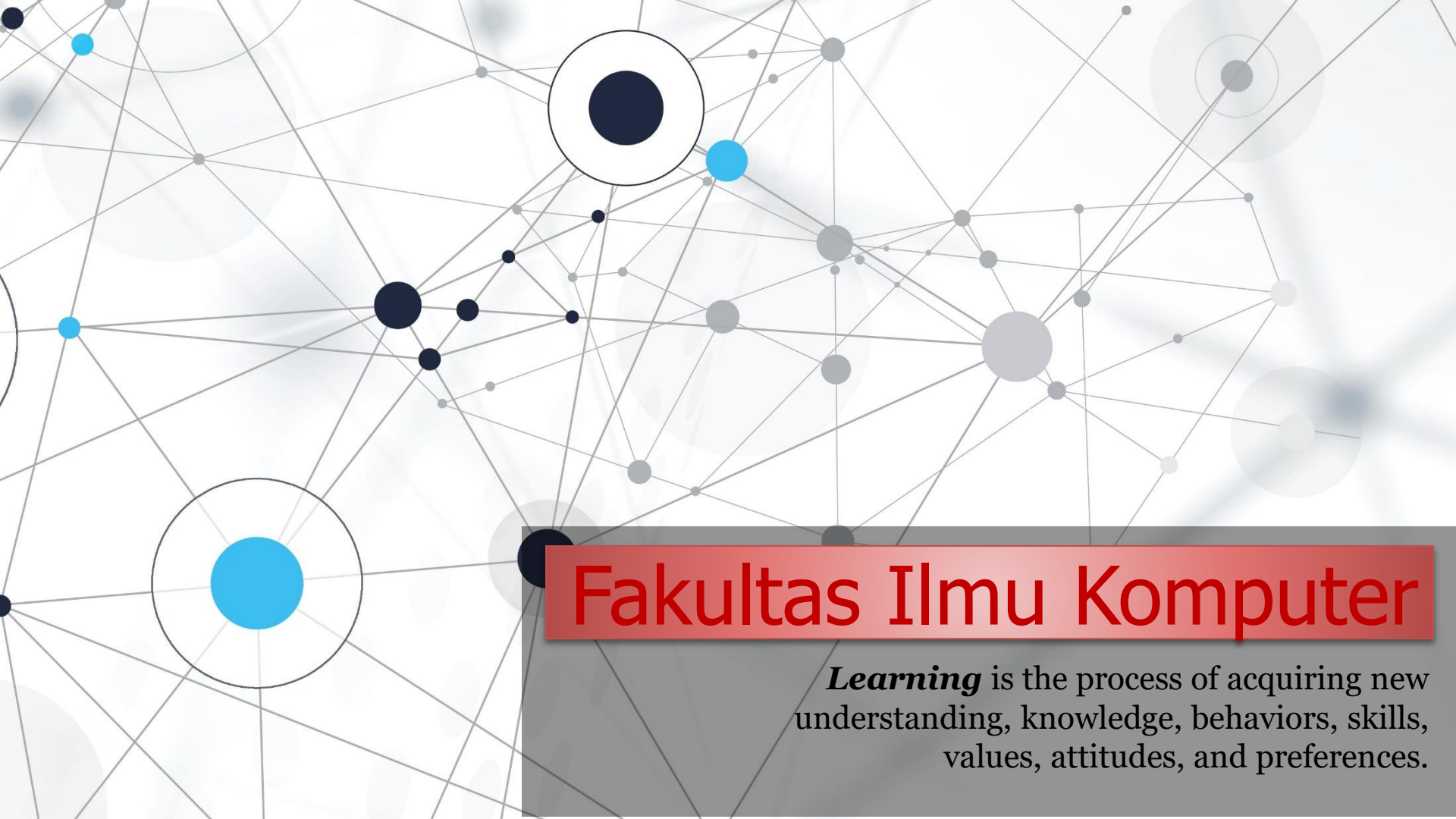




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Introduction to

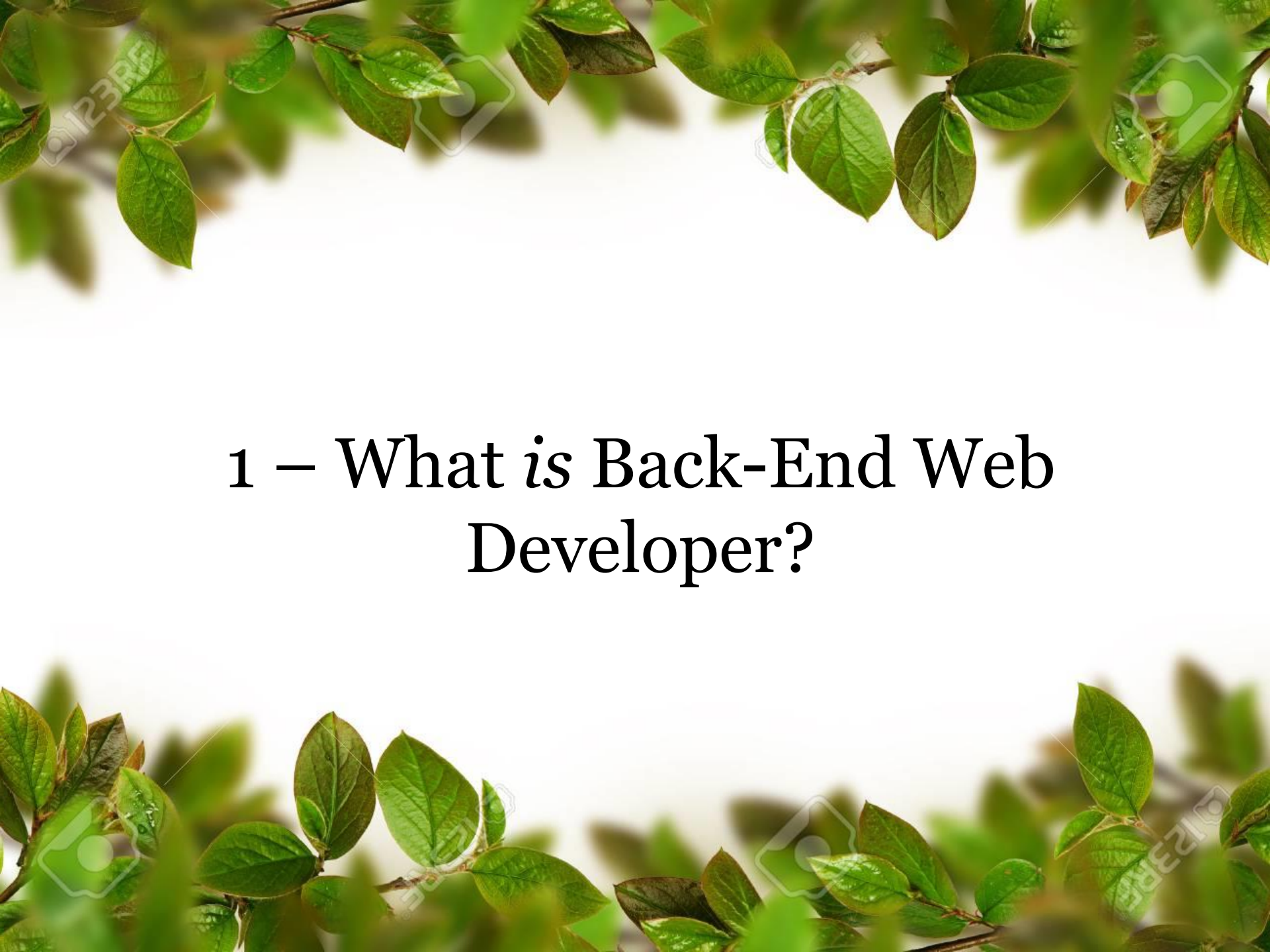
Back-End Web Development

MVC (Model View Controller) Design Pattern



Back-End Web Development

CMS *vs* MVC Frameworks



1 – What is Back-End Web Developer?

HTTP
Hypertext
Transfer Protocol.



Helen

HTTP

http://www.example.com

password: abc123



Without password encryption

Hacker see "abc123"

HTTPS
Hypertext Transfer
Protocol Secure.



Carol

HTTPS

https://www.example.com

password: abc123



With password encryption

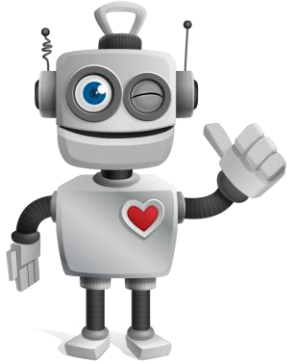
Hacker see "xyaerXzabc"

Apakah tugas seorang **back-end web developer** meliputi **keamanan** web untuk request (*permintaan*) HTTP dan HTTPS?



A Back-End web developer has the task of developing and maintaining components that are responsible for business logic, data processing, and interaction (*communication*) between servers and users (*end users* and *clients*) on a website or web application.

What is back-end development?



Back end development (often called the “**server-side**” development), is the creation of everything that goes on behind the scenes of a website or application that the user can’t see.



Back-end developers are responsible for building the technologies that host these database interactions and user requests. Back-end developers ensure the smooth and efficient functioning of a website or an application.



A back-end developer creates and includes things like business logic, calculations, database interactions, and performance in the server-side of an application or webpage.

Jenis-Jenis Web Development

FRONT-END WEB DEVELOPMENT

Tugas utama:

- ❖ Para developer pada bagian ini bertugas untuk membangun UI yang dapat membantu pengguna (*users*) untuk mencapai tujuan mereka di dalam situs.
- ❖ **Front-end web developer** juga sering ikut dalam aspek perancangan User Experience (UX) dalam project.
- ❖ Memiliki *background* di bidang UX dapat menjadi nilai plus bagi seorang front-end developer.



BACK-END WEB DEVELOPMENT

Tugas utama:

- ❖ Back-end developer bertugas untuk **mengelola server situs web**, program (*database*), dan software, dengan tujuan agar semua fitur tersebut dapat bekerja dengan baik.
- ❖ **Back-end developer** bekerja dalam sistem seperti ***server***, *sistem operasi*, **API**, dan *basis data*.
- ❖ Bertugas untuk mengelola *code* dalam situs untuk **keperluan keamanan**, *konten*, dan *struktur situs*.



What is a Backend Developer? Without “Web”?



Front End

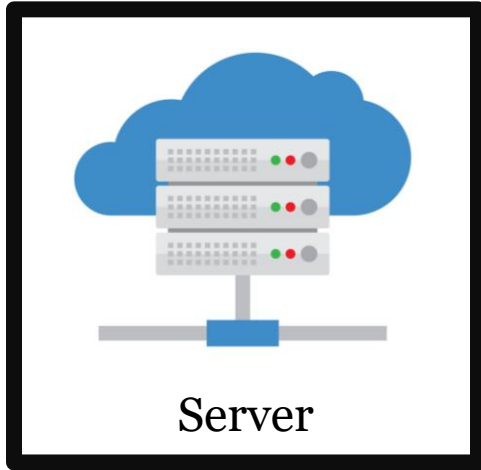
- Markup and web languages such as HTML, CSS and Javascript
- Asynchronous requests and Ajax
- Specialized web editing software
- Image editing
- Accessibility
- Cross-browser issues
- Search engine optimisation



Back End

- Programming and scripting such as Python, Ruby and/or Perl
- Server architecture
- Database administration
- Scalability
- Security
- Data transformation
- Backup

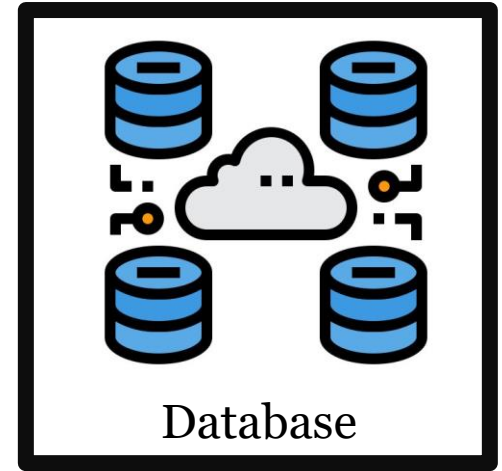
Skills & Tools Required *for* Back-End



Server

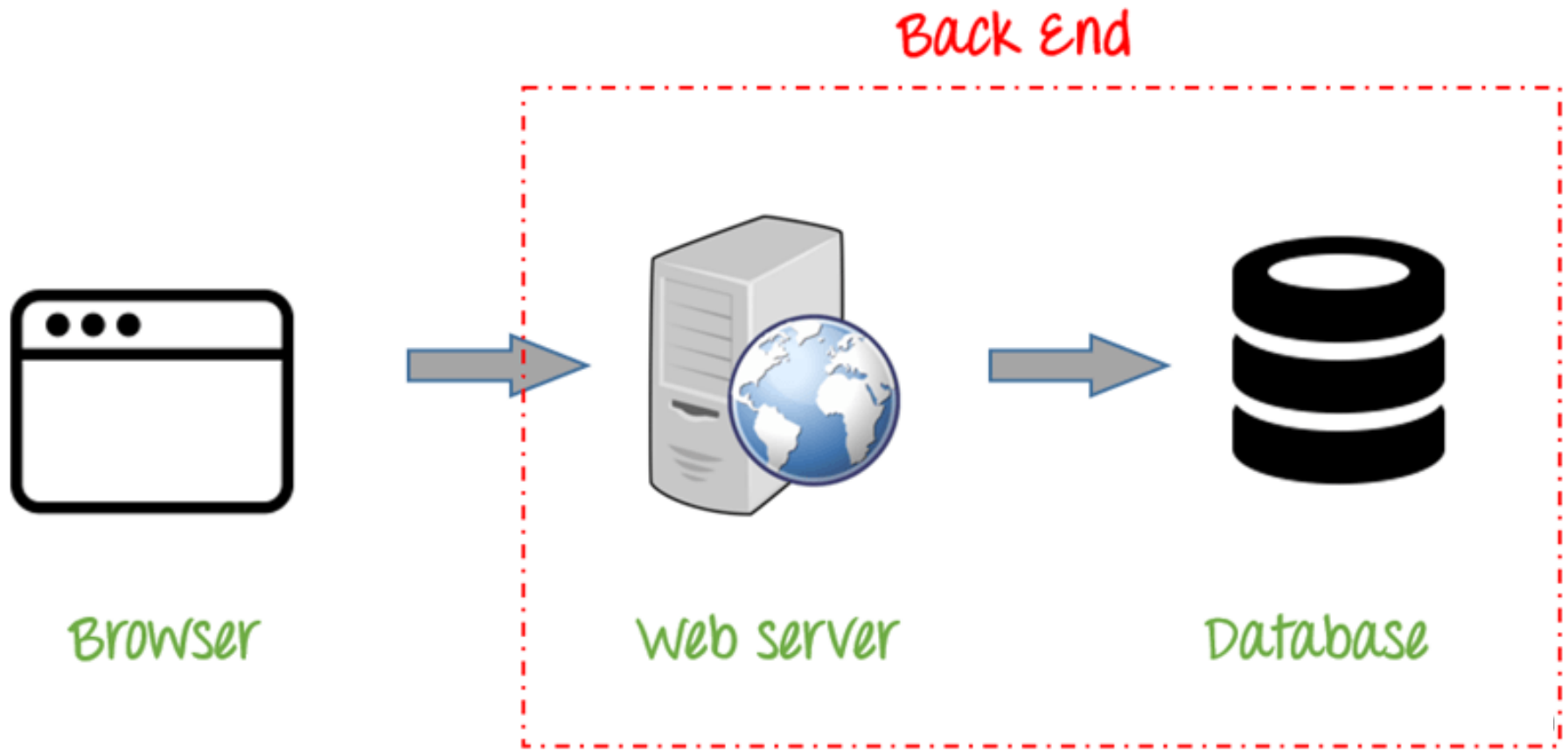


Application

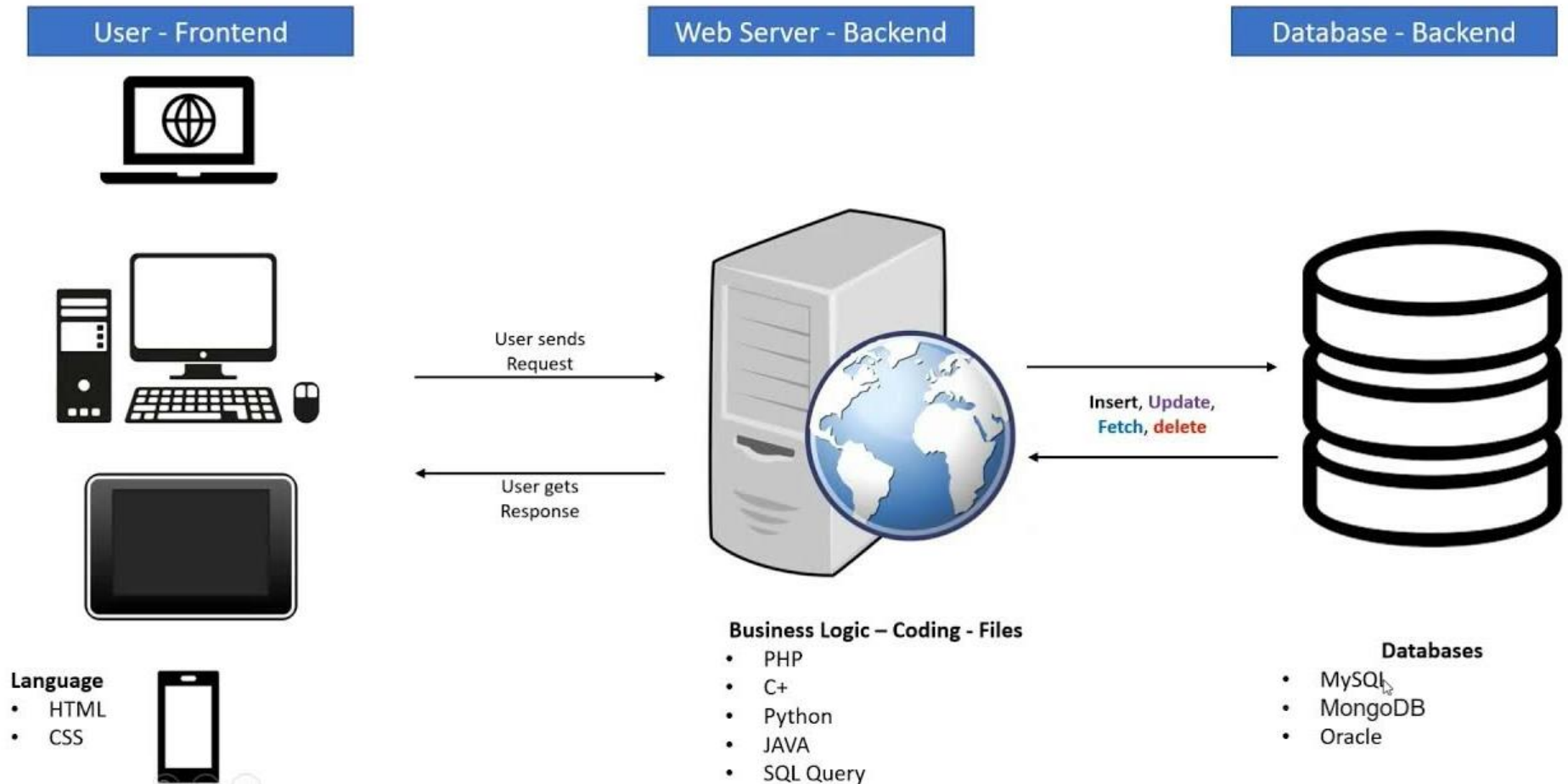


Database

In order to make the server, application and database communicate with each other, back-end developers use server-side languages like **PHP**, Ruby, **Python**, Java and .Net to build an application and tools like MySQL, Oracle and SQL Server to find, save or change data and serve it back to the *user* in front-end code.



End-users/client do not directly access the backend of an application or computer system. Technology like servers, databases, and APIs allows features such as a user interface (UI) to work. Backend developers responsible to build and to maintain those technologies.



Backend developers use programming and scripting languages like Python, Ruby, Java, .Net and PHP. They focus on concerns like whether the site is secure or how data will be managed.

Skills & Tools Required *for* Back-End



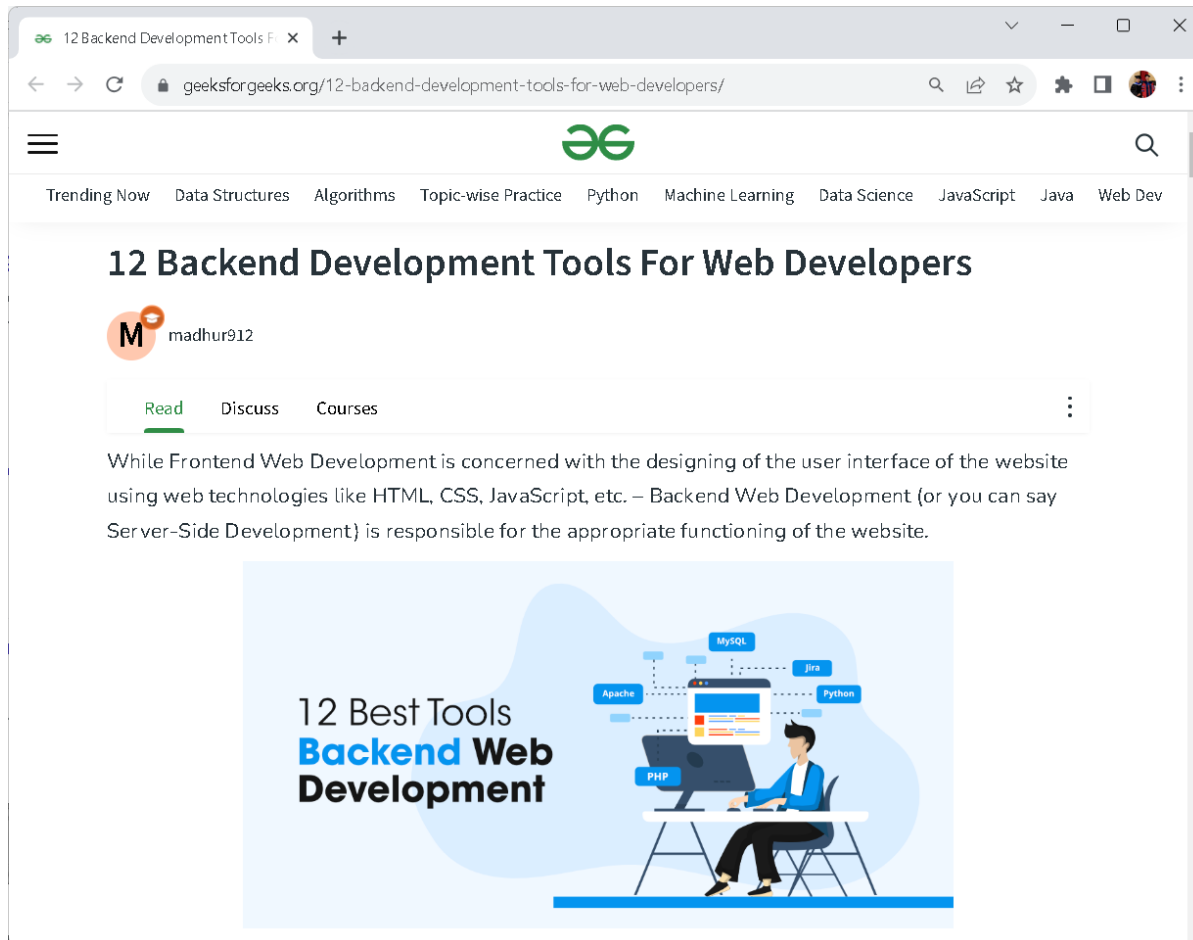
Back-end developers ensure the website performs correctly, focusing on databases, back-end logic, application programming interface (APIs), architecture and servers.

Back-end developers must be familiar with many kinds of tools and frameworks.

The back-end of a website consists of a server an application, and a database. Job openings for back-end developers often also call for experience with **PHP frameworks** like Zend, Symfony, CakePHP, *etc.*

Things can become a bit convenient and easier for the Backend Developers if they choose the right set of backend development tools available in the tech world.

- 1) What is the type of **server-side scripting language** (PHP, Python, Perl) you want to use?
- 2) What popular, free and **open-source databases** are considered (MySQL and MongoDB)?
- 3) How about open-source cross-platform **web server** (Apache and NGINX)? and
- 4) Other tools may also help you (**IDCloudHost**, **Postman** and Docker containerization platform allows to create, deploy and run applications)?



Back-end developers harus terbiasa dengan banyak jenis programming languages, tools dan juga frameworks.

Bagaimana cara agar bisa tahu dan mengerti banyak bahasa pemrograman dan juga frameworks? Apakah ini membutuhkan banyak waktu juga di pelajari satu per satu?



A decorative border of green leaves and branches at the top of the slide.

2 - CMS *vs* MVC Frameworks

A decorative border of green leaves and branches at the bottom of the slide.

Requirements



- ☐ You should already be familiar with PHP basics.
- ☐ You should have *web server* and MySQL installed on your own operating system or hosting. If you haven't – you can find any tutorials from Youtube.
- ☐ You should have IDE installed. We recommend **PHP Storm** but **Notepad++** is also affordable.

Description

1. Learn MVC by creating our own PHP framework step by step.

- i. MVC pattern, which stands for Model View Controller, is a widely – used programming technology and the most popular technique in web applications development.
- ii. The most popular frameworks such as Laravel, Symphony, Yii, Zend are based on MVC pattern. So, it undoubtedly worth learning.

2. The best way of learning it is creating your own MVC framework and CMS.

- i. Step by step we'll be creating MVC web application. Further the result of application developing may be put into practice.
- ii. As a consequence, we will create a light-weight framework and CMS which may be used in your project. It would contain pages view and editing, contact form, admin section and authorization.
- iii. We will learn the best practices in PHP OOP, sessions and MySQL interaction.
- iv. Basic knowledge of PHP OOP is all that you need.

What is a CMS?

- ❑ **CMS** stands for **Content Management System**, that *reduces the role of the developer* and it is more of tool knowledge and less of code that **provides default templates** and **themes for building the websites**.
- ❑ CMS is a base premade system that you can extend with modules or plugins. This provides a number of web designs and plugins for mostly all general problems.
- ❑ The developer has to only customize the tools (**web templates and plugins**) to build a website.
- ❑ Although the developers can write their own plugins and templates but in most of the cases they need not this work. You really don't need to know any code to use a CMS.
- ❑ Examples for CMS are **wordpress, joomla, umbraco** etc.

What is a CMS?

People also ask

What is a CMS website template? ^

A **CMS** is a tool used to manage and modify the content of a **website** without having to write program code. A **CMS** allows you to edit, modify, organize, delete, publish the **website** content with the help of predefined **templates**, themes and plugins. Examples of the most popular **CMS** are WordPress, Joomla and Drupal.

<https://www.pickaweb.co.uk> > how-to-build-websites-or-b...

How to build Websites or Blogs using Templates and CMS -
Pickaweb

Search for: [What is a CMS website template?](#)

What is Moto CMS templates? v

What is a CMS theme? v

Which website template is best? v

Is GoDaddy website builder a CMS? v

What template means? v

What is a CMS?

cms web template - Google Search

google.com/search?q=cms+web+template&sa=strict&hl=en&source=Inms&tbn=isch&sa=X&tved=2ahUKewjv6ZfHsOvwAhWZ63MBHVfoAIUQ_AUoAXoECAEQAw&biw=...

Google

cms web template

All Images News Videos Shopping More Settings Tools Collections SafeSearch on

bootstrap admin dashboard django cms themes classified ecommerce classified ads product

CMS Templates from ThemeForest
themeforest.net

Random CMS Templates from ThemeForest
themeforest.net

Cms free website templates for free ...
all-free-download.com

22+ Admin CMS Themes & Templates | Free ...
template.net

CMS Templates from ThemeForest
themeforest.net

22+ Admin CMS Theme...
template.net

Free Control Panel Website Templates ...
free-css.com

9 new Webflow CMS templates | Webflow Blog
webflow.com

22+ Admin CMS Themes & Templates | Free ...
template.net

www.kubzon.org

NeueAdmin

Best Marketing

Advantages of CMS



- ❑ **Ready-to-use web templates**, i.e., not need to write code. Just customize and present website without any hard afford.



- ❑ **Ready-to-use plugins**. You have to only search for plugin for any functionality. For example **wordpress** has more than 17,000 plugins.



- ❑ **CMS is good for having many websites with typical functionality.**



- ❑ With CMS, there's no need to reinvent commonly used stuff such as *breadcrumbs (alat navigasi pada website untuk memudahkan pengunjung setelah menu utama pada header website), easy media files, menu management, etc.*

A **plugin** is a piece of software that acts as an add-on to a **web** browser and gives the browser additional functionality. **Plugins** can allow a **web** browser to display additional content it was not originally designed to display.

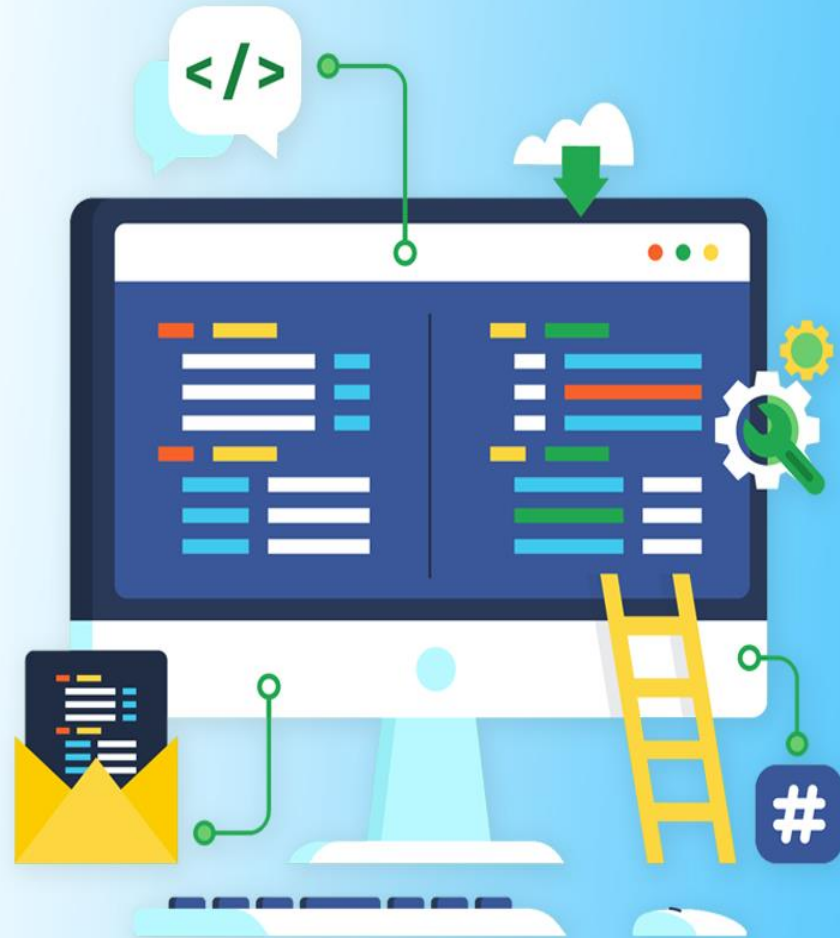
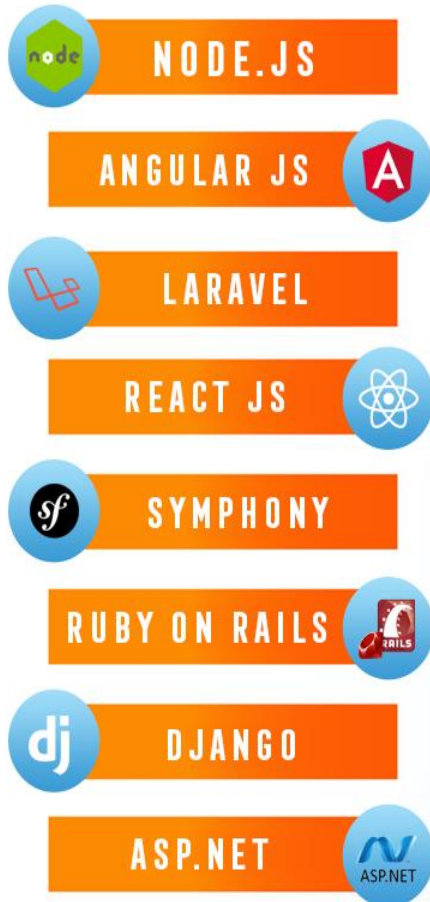
Disadvantages of CMS

- ❑ CMS **restricts what you can do**. CMS are for some type of general purpose like wordpress is for blogging. So it is not possible/easy (if possible) to build a chatting and video calling website using wordpress.
- ❑ Slow customizing for unique requirements (***if you have not build your own CMS***). CMS is build for all users so customizing it for some specific purpose (*which is not usually common*) is hard.
- ❑ **Changing the code can be hard and, worse, risky** because you might lose compatibility with new versions of the CMS or other plugins.
- ❑ With CMS it's impossible to use good programming practices such as automated tests and refactoring (**black box testing**).
- ❑ With CMS, if a performance problem is in the core, you can't fix it. Although this is not a common problem.

What is MVC framework?

- ❑ Whereas MVC stands for **Model View Controller**, it is a pattern that decouples the view (UI) and Controller logic.
- ❑ It is a software architecture, that you can use to develop application.
- ❑ It can help to make the best use of the **HTML, CSS, javascript** and **AJAX** features to give a *highly dynamic web applications*.
- ❑ MVC is widely used in **ASP .NET, PHP, Python** and **Java**. To use any MVC you must have a good knowledge of that language as well as the libraries provided by that MVC.
- ❑ You have to write all **plugins** and **templates** yourself.

TOP **WEB** DEVELOPMENT FRAMEWORKS



Advantages of MVC



- ☐ Full control. You have no restriction as you can do any thing.



- ☐ Good inbuilt API for all types applications.



- ☐ MVC is good for very custom websites with unique functionalities.



- ☐ Maintenance is easy.



- ☐ Easy to ***change code*** as the code does not depends upon the third party.

Framework MVC

Bahasa	Framework
PHP	CodeIgniter, Laravel, Yii, dll
Java	Spring MVC, JSF, Struts, dll
Python	Django, CherryPy, dll
Ruby	Ruby on Rails, Sinatra ,dll
Javascript	AngularJS, React, Backbone.js, dll

Disadvantages of MVC



☐ *Require high knowledge of programming languages as you have to write code.*



☐ Require knowledge of inbuilt APIs.



☐ Not required for small websites (*like blogging websites*)



☐ Take much time to build a website.

More Comparison

- ☐ CMS is expensive to set up and maintain whereas MVC is expensive to set up but not in the long run.
- ☐ CMS is fast than MVC.
- ☐ MVC is best for dynamic websites in long run rather than CMS.



The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some visible veins and slight shadows, creating a natural border around the central text.

3 - More about MVC

MVC

- **Model, View, Controller**
- **Object Oriented**
 - Is communication between Model, View, and Controller *Objects*
- **Software Engineering**
- **Organization**
- **RAD**
 - Code re-use.

MVC Is Not New!

- Smalltalk-80 (circa **1980**)
 - MVC first discussed in 1979 by Trygve Mikkjel Heyerdahl Reenskaug
- MFC (Document/View)
- Java's Swing
- QT4 (KDE)
- Apple's Cocoa (CoreData)
- Notice a trend?
 - MVC is perfectly suited for GUI development.

MVC and the Web

- ❑ Made popular by Ruby on Rails
 - ❖ A good number of PHP MVC frameworks are Rails inspired
- ❑ Wikipedia currently lists about 17 PHP MVC frameworks
 - ❖ CakePHP
 - ❖ Symfony
 - ❖ CodeIgniter
 - ❖ Zend Framework
 - ❖ Make your own (we will try to do !!!)

MVC - Model

- Wikipedia:
 - “The domain-specific representation of the information that the application operates.”
- **Data Storage/Access (OOP)**
 - Often Data sourcebacked
 - MySQL, MSSQL, Web Service, it doesn't matter...
 - Abstraction, Abstraction, Abstraction!

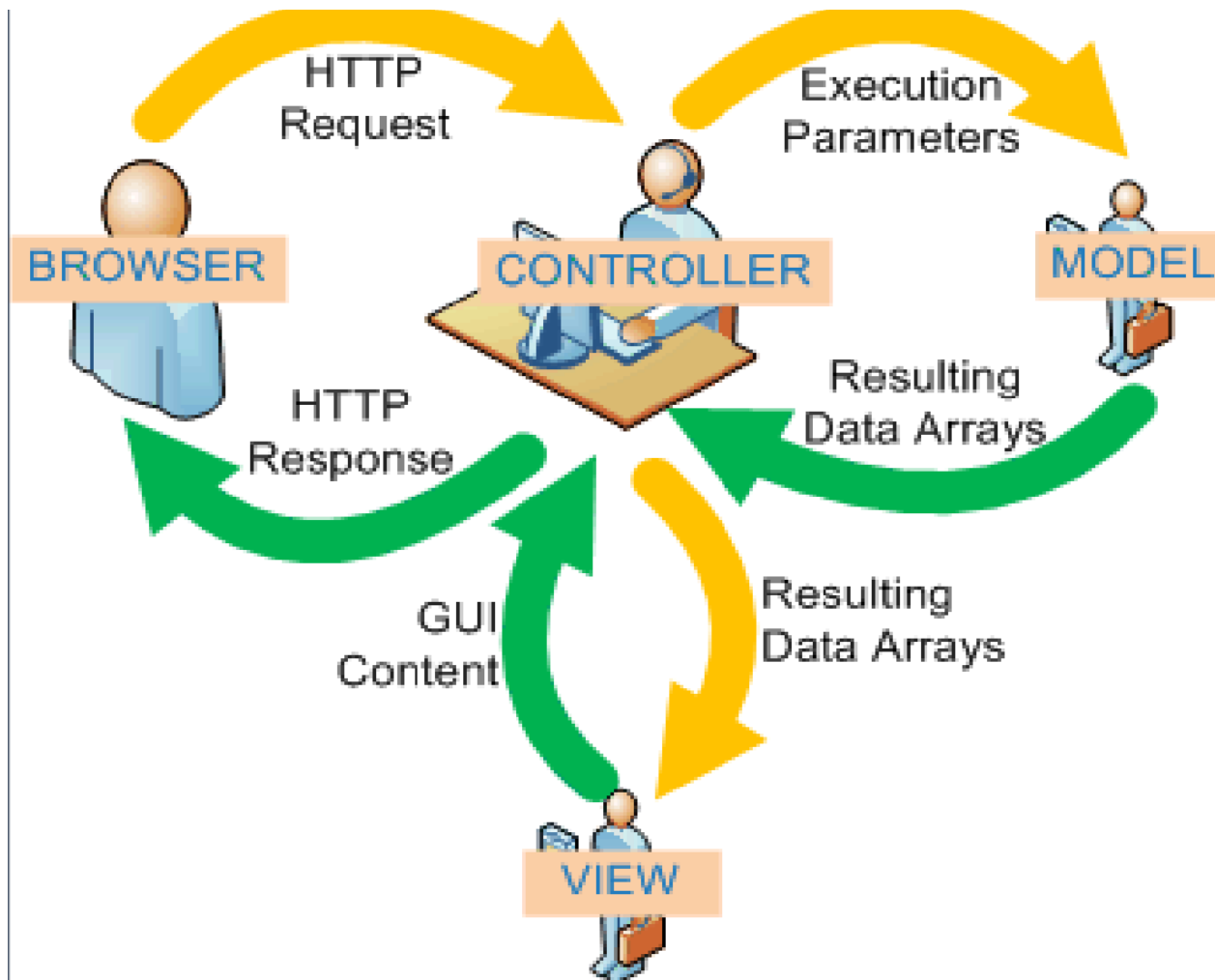
MVC - View

- Wikipedia:
 - “Renders the model into a form suitable for interaction, typically a user interface element. Multiple views can exist for a single model for different purposes.”
- HTML Templates
 - All formatting related code belongs here
 - A Smarty object is a good example
 - Any template engine works, however...
 - Abstraction, Abstraction, Abstraction!

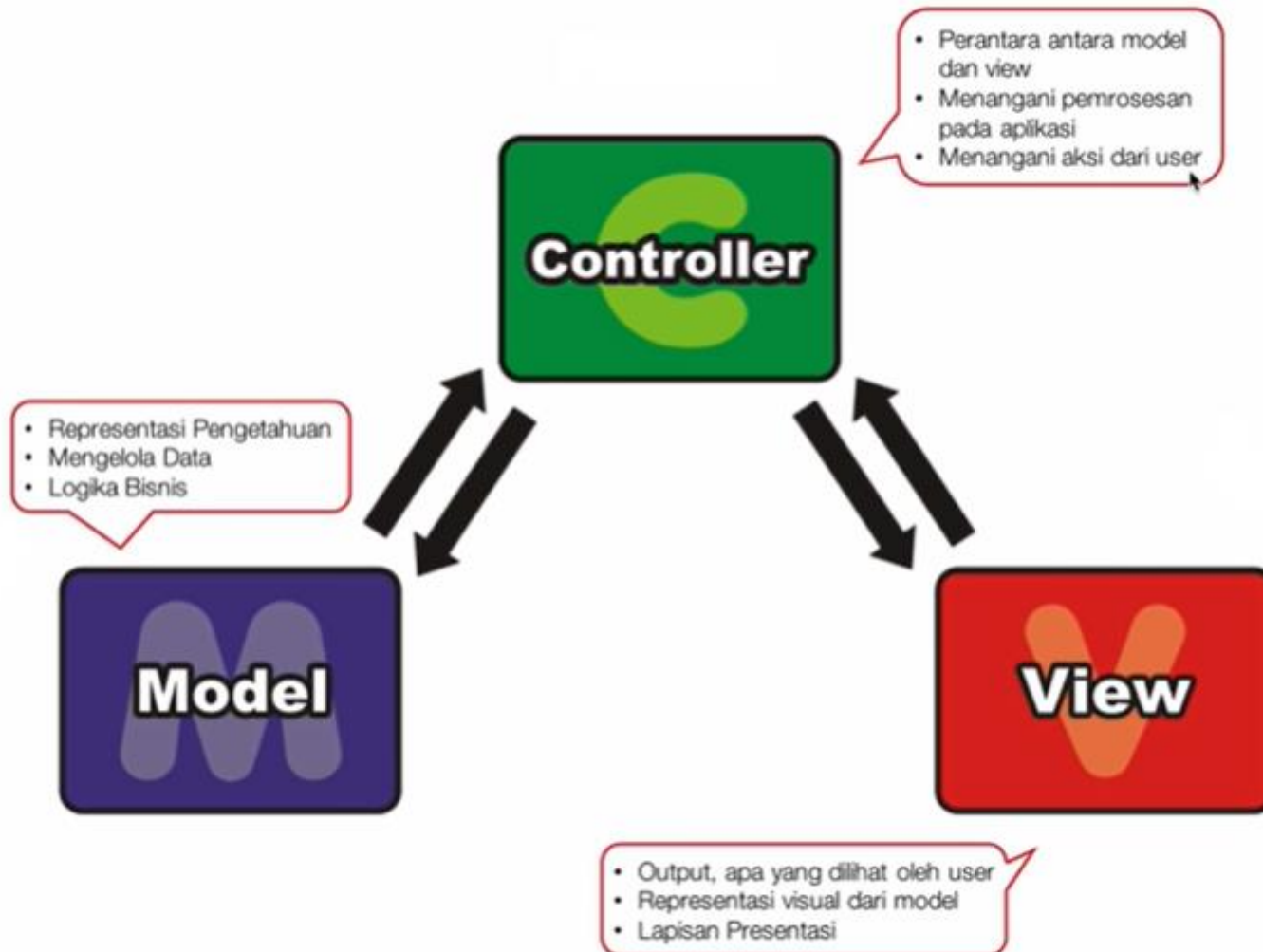
MVC - Controller

- Wikipedia:
 - “Processes and responds to events, typically user actions, and may invoke changes on the model.”
- A BIG DEAL
 - **Process user inputs, communicate with Models and Views**
 - The Go-Between
 - Much of the application’s ***core logic***
 - Utilize the abstraction of before!
 - **Invoke model, assign values to views**

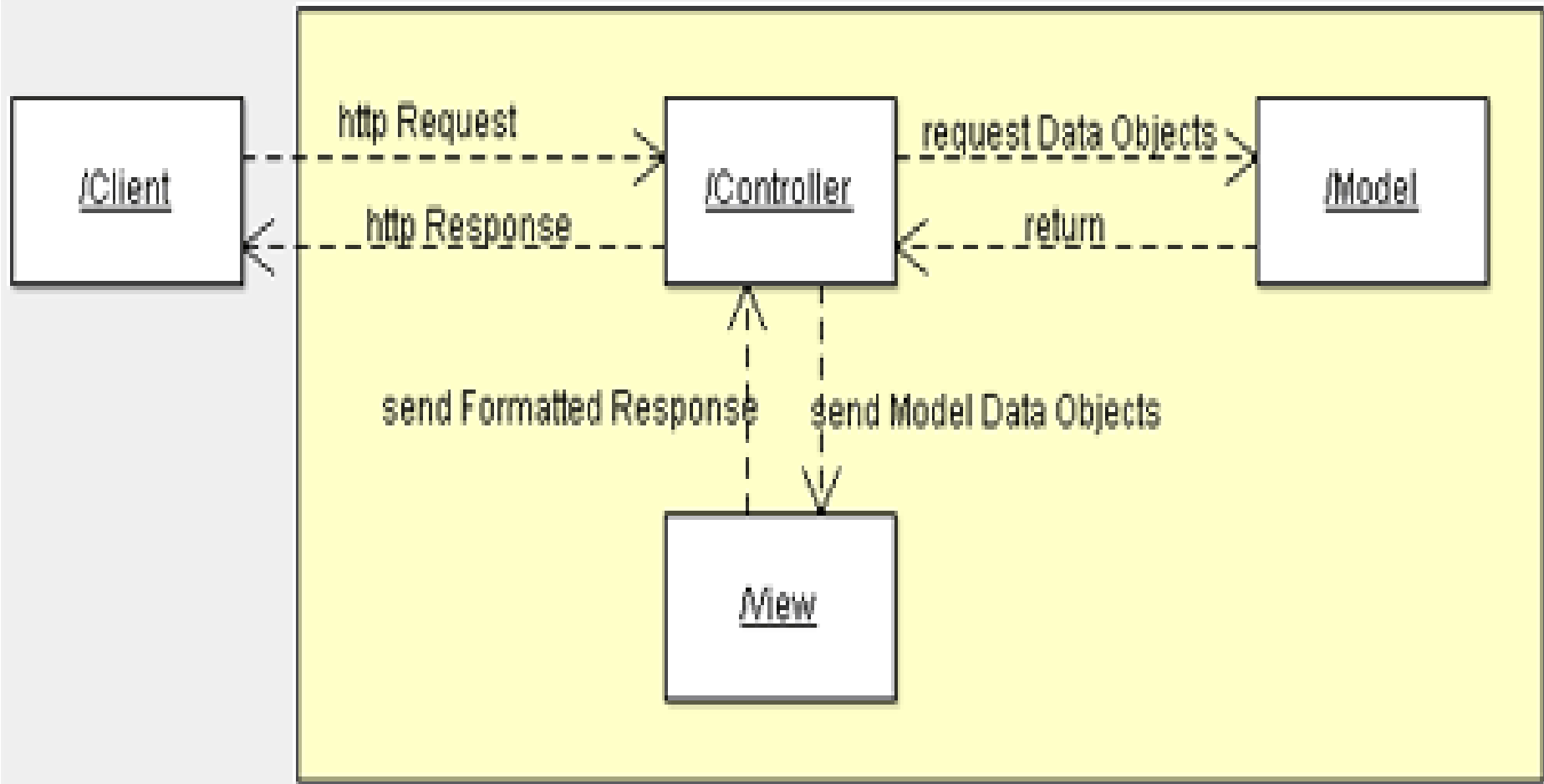
MVC - Putting It Together...



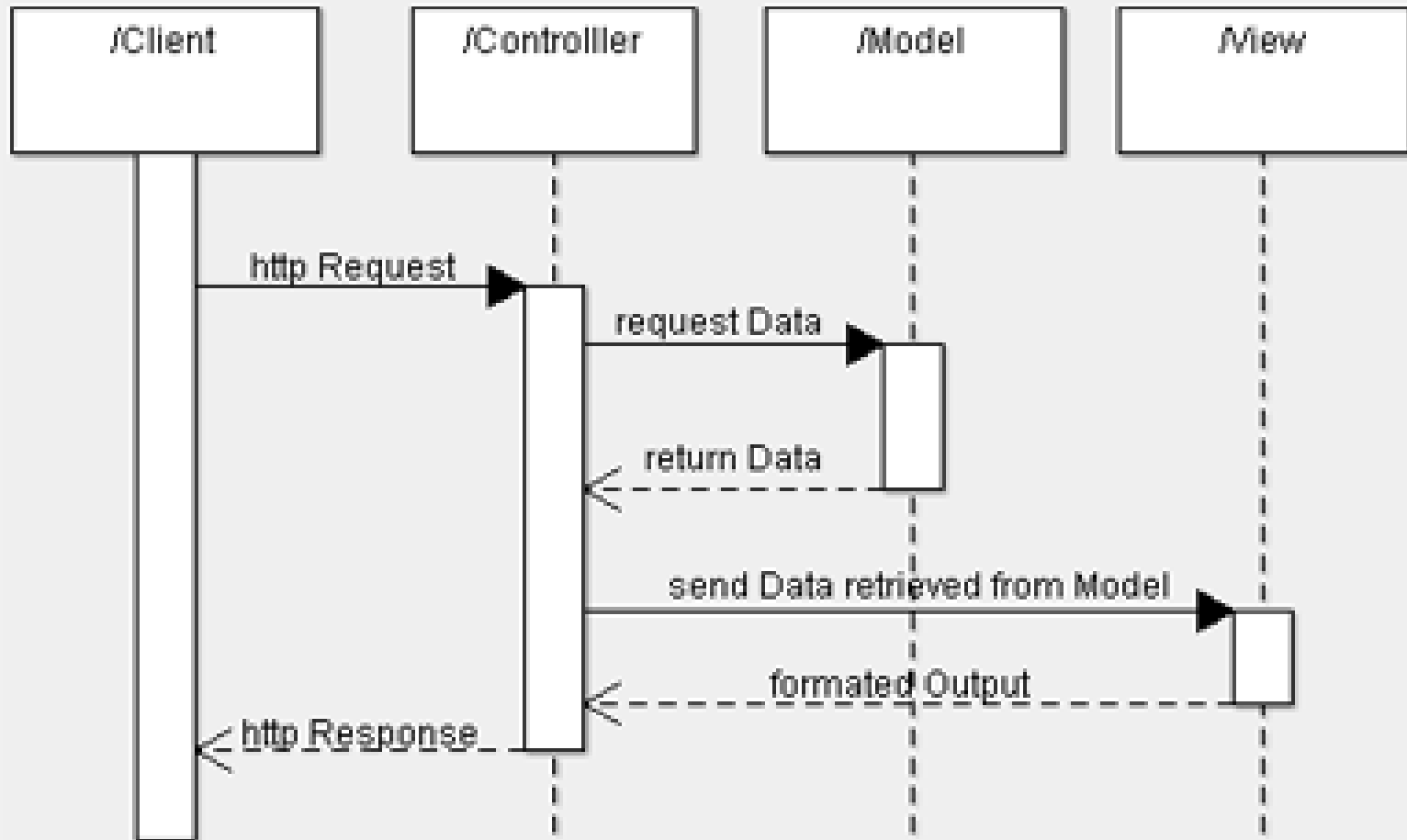
MVC - Putting It Together...



MVC Collaboration Diagram

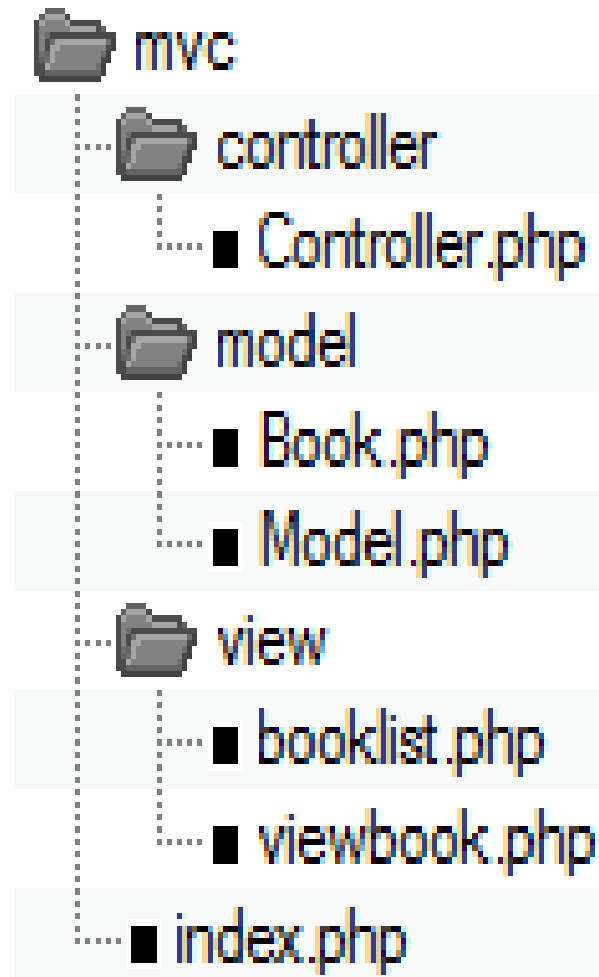


MVC Sequence Diagram



Common Traits/Characteristic

- **index.php**
 - Parses the URL
 - Often mod_rewrite used:
`/controller/method/arg0/value0/arg1/.../valuen`
 - Initializes proper controller
 - Executes correct method
 - **Facilitates communication from controller to view**
 - Signals view to render



Directory Structure

- Outside of web root. More secure! (*generally*)
- config/
- controllers/
- models/
- views/
 - Usually not PHP files
- includes/
 - Any extra libraries, e.g. PHP Mailer
- framework/
 - Framework system files (do not have to reside in application)

Helpers and Plug-ins

- Usually common functions used by Views
 - ▣ E.g. `format_phone()`, `create_calendar()`, etc.
- Some loaded through **index.php**, some loaded specifically by **controller**
- Do not allow direct manipulation of models or controllers
 - ▣ Any code requiring access to models should be in the controller, not view

Why MVC?

- **Good architectural design**
 - Code is organized and structure
 - Code structure lends itself to an easy to understand directory structure
- **Easy code maintenance**
 - Because of abstraction, better strategic positioning of code will minimize the hunt for places to change
- **Easy to extend and grow**
 - Modify parent classes, drop in new controller, *etc.*

Sites Using PHP MVC Frameworks

- CakePHP
 - MozillaAddons
 - The Onion's Online Store
- Symfony
 - TED.com
 - Yahoo! Bookmarks
- Pretty much everyone.



VS



Sudah siapkah kalian untuk menjadi seorang Back-End Web Developer? Skill apa saja yang sudah kalian punya?

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

